

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS

COURSE CODE: CSA0706

Course Outcome CO1: Apply the functions of the OSI layer in enabling reliable data transmission across networks.

Code of Conduct:

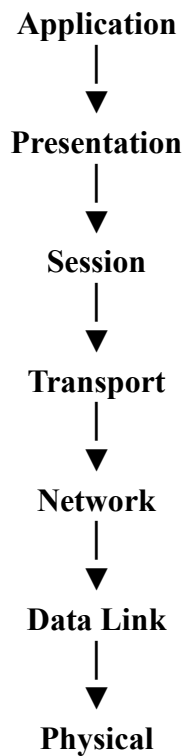
I _____ **M Nishanth (192511154)** _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M Nishanth

Annexure A

CONCEPT MAPPING

1. Concept Map



Missing Layers

- Session Layer
- Data Link Layer

Explanation:

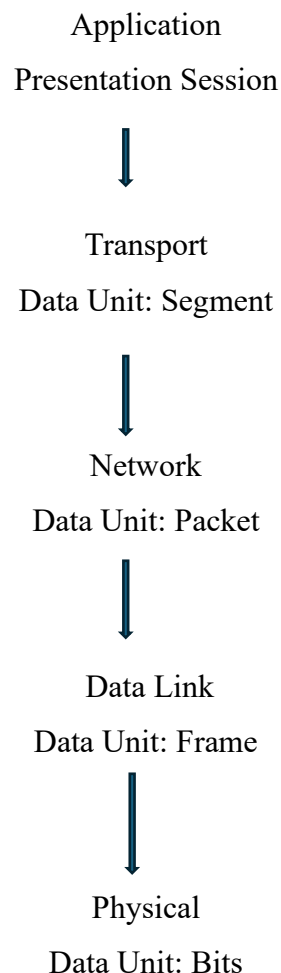
The **Session Layer** (Layer 5) establishes, manages, and terminates communication sessions between applications. It provides synchronization using checkpoints so communication can continue from the last saved point if interrupted. It also controls dialog management, ensuring orderly data exchange.

The **Data Link Layer** (Layer 2) provides reliable node-to-node communication. It performs framing, physical addressing (MAC addresses), error detection using CRC, flow control, and media access control. It ensures that frames reach the next device without transmission errors.

Analysis:

Reliable communication depends on every OSI layer working together. The Session Layer maintains continuous communication between applications, while the Data Link Layer ensures reliable transfer over the physical medium. Without these layers, communication would suffer from session failures, frame errors, and unreliable transmission.

2. Concept Map



Explanation

Each OSI layer converts the data into a different **Protocol Data Unit (PDU)** before sending it to the next layer. This process is called **Encapsulation**.

1 Transport Layer → Segment

- Divides application data into smaller units called **segments**.
- Adds **port numbers, sequence numbers, and error-control information** to ensure reliable communication.

2 Network Layer → Packet

- Encapsulates each segment into a **packet**.
- Adds the **source and destination IP addresses** so routers can forward the data to the correct network.

3 Data Link Layer → Frame

- Encapsulates the packet into a **frame**.
- Adds the **source and destination MAC addresses** and an **error detection field (CRC)** for reliable communication between directly connected devices.

4 Physical Layer → Bits

- Converts the frame into **bits (0s and 1s)**.
- Transmits these bits through the communication medium (copper cable, fiber optic cable, or wireless signals).

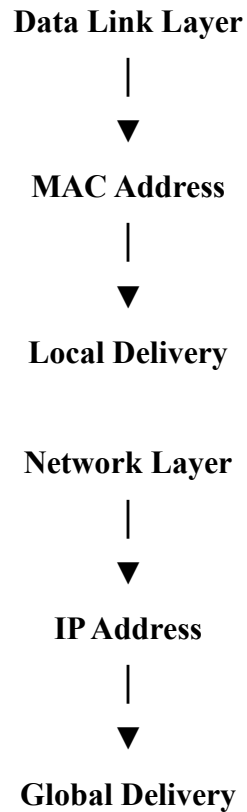
Analysis

During encapsulation:

1. Application generates data.
2. Transport converts data into Segments.
3. Network converts Segments into Packets.
4. Data Link converts Packets into Frames.
5. Physical converts Frames into Bits for transmission.

At the receiver, decapsulation reverses these steps, reconstructing the original application data accurately.

3. Concept Map



Explanation:

The **MAC Address** is a unique hardware address used only within a local network (LAN). Switches use MAC addresses to forward frames between devices connected to the same network. The **IP Address** uniquely identifies devices across different networks. Routers use IP addresses to determine the best path for forwarding packets over the Internet.

Analysis:

Data transmission requires both addressing methods:

MAC Address → delivers data within a local network.

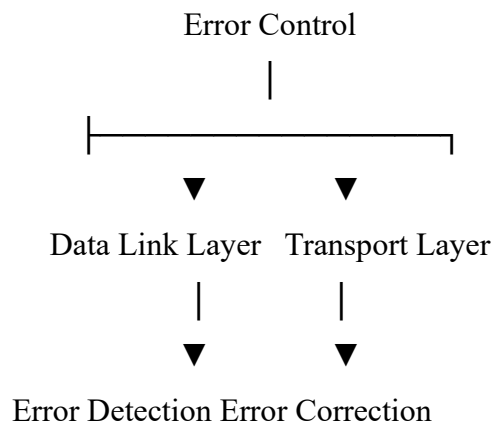
IP Address → delivers data across different networks.

When sending data to another network:

1. IP determines the destination network.
2. Routers forward the packet.
3. Each local network uses MAC addresses to deliver the frame to the next device.

Together, MAC and IP addressing ensure complete end-to-end communication.

4. Concept Map



Explanation:

The **Data Link Layer** detects transmission errors between directly connected devices using techniques such as CRC (Cyclic Redundancy Check). If an error is detected, the damaged frame is discarded or retransmitted.

The **Transport Layer**, especially TCP, provides end-to-end reliability. It detects missing segments through acknowledgements, sequence numbers, timers, and retransmissions.

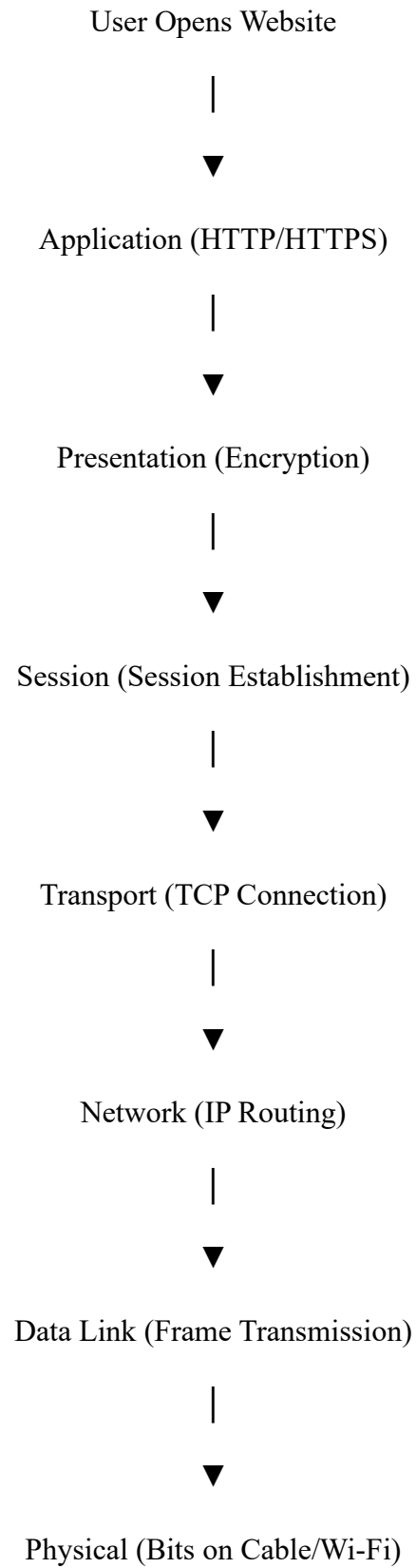
Analysis:

Both layers cooperate to improve reliability.

- Data Link Layer prevents corrupted frames from moving further.
- Transport Layer guarantees complete delivery between source and destination.

This layered approach minimizes data corruption and ensures accurate communication.

5. Concept Map



Explanation:

During web browsing:

- Application Layer uses HTTP/HTTPS.
- Presentation Layer encrypts data using SSL/TLS.
- Session Layer manages communication sessions.
- Transport Layer establishes TCP connections and ensures reliable delivery.
- Network Layer routes packets using IP.
- Data Link Layer transfers frames using MAC addresses.
- Physical Layer transmits binary bits over the communication medium.

Analysis:

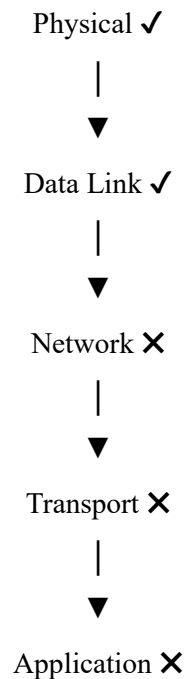
Failure in any layer affects the entire communication process.

Examples:

- Physical Layer failure → No Internet connection.
- Data Link failure → Frame transmission errors.
- Network Layer failure → Packets cannot reach destination.
- Transport Layer failure → TCP connection cannot be established.
- Application Layer failure → Website cannot load.

Since each layer depends on services from the layer below, a failure propagates upward and disrupts web browsing.

6. Concept Map



Analysis :

The concept map indicates that:

- Physical Layer is functioning correctly.
- Data Link Layer is also functioning correctly.
- Network Layer has failed.

Because the Network Layer is responsible for IP addressing and routing, packets cannot move beyond the local network.

As a result:

- Transport Layer cannot establish end-to-end communication.
- Application Layer cannot access services such as web browsing, email, or file transfer.

Troubleshooting Steps

1. Verify IP address configuration.
2. Check subnet mask.
3. Verify default gateway.
4. Test connectivity using ping.
5. Check router status.
6. Verify routing tables.
7. Ensure DNS configuration is correct.

Conclusion

The concept map quickly identifies the failed layer, allowing network engineers to troubleshoot systematically instead of checking every component. Since the Physical and Data Link layers are operational, the issue is isolated to the Network Layer, reducing troubleshooting time and improving efficiency.